

# Synoptic: Query-Driven Multimodal Product Review Summarization System

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## Abstract

Product reviews play a critical role in making purchase decisions. A recent shift toward multimodal reviews highlights the preference for visual and audio information over texts. While long-form review videos are information-rich, viewers often skip to find what they need, resulting in missing information. Moreover, videos are difficult to search for specific content, leading to ineffective information gathering. In this work, we surveyed 177 people to learn about the challenges and strategies to gather information from product review videos before making purchase decisions. Based on our findings, we designed and developed Synoptic, an interactive multimodal video player that allows users to search within videos and generate video summaries based on customized queries. Our initial evaluation with 40 participants suggests that Synoptic helped viewers gather personalized information from product review videos effectively and efficiently. It also enables purchase decision-making while maintaining user agency and prioritizing their preferences.

## CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**.

## Keywords

Product review videos, video summarization, interactive transcripts, multimodal

### ACM Reference Format:

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## 1 Introduction

Consumers often consult product reviews to assess the product's quality, performance, durability, and other attributes to inform their purchase decisions [1, 23, 30, 48]. These reviews are predominantly textual, with occasional images. While beneficial, they often lack comprehensive visual demonstration of products that might impact

purchase decisions [36]. Recently, there has been a shift in preference towards multimodal product reviews [3] containing video, audio, and text. This transition has been accelerated by advancements in social media and multimedia platforms that have shifted consumers' affinity towards multimedia content; making video reviews the preferred method for consumers seeking information-rich product reviews [32]. Prior research suggests that 85% of consumers prefer video reviews over textual ones to gather more comprehensive information before making purchase decisions [13, 17]. However, these product review videos often contain content that may not be sought after by all viewers. For instance, some consumers may be interested in a product's specifications, while others may prioritize aesthetics or usability [38, 53]. Long-form videos (over 10 minutes [39]) deliver rich content but people seeking targeted information often get lost in the details. While text reviews can be searched for relevant content easily [34], exploring and searching video content is non-trivial [5, 26].

While existing approaches investigated timestamps [37], subtitles [42], and chapters [55] to enable video exploration, these author-generated markers often fall short of addressing viewers' preferences for customized content [29, 42, 55]. Some existing approaches enable searching for specific content through transcription [56] with limited success as the visuals may not match the uttered speech. Others have explored interactive technologies employing both crowd-sourcing and machine-learning methods to develop navigation features for video exploration [4, 6, 8, 44]. In addition, video summarization has been explored as an alternative to reduce information-gathering efforts. For instance, Barthel et al. proposed a graph-based summarization method by extracting key frames from the video [9]. Similar approaches include both text-based and frame-based systems to generate meaningful video summaries to help identify essential product details more efficiently without the need to watch the entire video [28, 41, 46, 54]. However, these methods do not support custom queries where users can search the video with specific keywords or questions to retrieve contextual video segments of interest [50]. Existing query-based summarization approaches generate summary videos that are a collection of static image frames [52] or textual summary [35] and not videos by themselves. The lack of an effective way to search for information in videos along with low attention span [14] often results in frustration, dissatisfaction [5] and insufficient information to make confident purchase decisions [30] based on product review videos. Effective alternatives to search for information in review videos remain an under-explored research challenge.

To better understand how product review videos impact people's purchase decisions, we conducted a formative study by surveying 177 people. We found that product review videos provide

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specifications and demonstration of functionalities, enabling more concrete and confident decisions. However, review video length impacts information gathering as long-form videos result in disengagement and short-form videos often lack sufficient information. Participants often skip or fast-forward long-form videos to navigate review videos in a short time. While doing so, they often miss valuable information. We also saw the need for summaries that can help them make quicker decisions.

Based on the survey responses, we designed and developed Synoptic, a video player that allows users to find information from product review videos using custom queries. We used a large language model (LLM) and a vision language transformer model (ViLT) to identify segments from the video that match the user query and combined them to create the summary videos. Synoptic has two views; the Custom Summary view provides a summary video based on users' query and the Overlay view presents the complete video, with highlights of the segments that are relevant to the user summary. Finally, we added an interactive transcript and relevant clips as alternative tools for navigating videos. With Synoptic, users can generate customized summary videos based on their information needs and reduce the time needed to gather information from long-form videos. We conducted an initial evaluation of Synoptic with 40 participants, where we asked them to use our system to watch product reviews and make purchase decisions. From the participants' feedback, we found that Synoptic enabled them to personalize and customize their product review video-watching experience, identify information efficiently, and make confident purchase decisions.

## 2 Formative Study

### 2.1 Participants and Procedure

We used Prolific [43] to recruit 180 participants from the US and UK, as they demonstrate similar purchase behaviors online [21]. We removed three participants for incomplete submissions. Among 177 participants, 79 used he/him, and 98 used she/her pronouns. Their average age was  $29.96 \pm 9.22$  years. Participants mentioned watching product review videos on several platforms, with YouTube being the dominant (167 participants), followed by TikTok 67 participants).

In the survey, we asked participants questions about their product review video-watching practices and preferences, their impact on purchase decisions, and the challenges they face. Participants spent  $9.63 \pm 7.42$  minutes on the survey and they were compensated with USD \$20/hour. We conducted the survey through Qualtrics and analyzed the responses using thematic analysis [10]. Two researchers separately performed open coding and consolidated them into a codebook with intercoder reliability of 0.90 (Krippendorff's alpha). Two researchers used the codebook to perform another coding pass. The research team then identified major themes from the coded data and finalized them over six sessions.

### 2.2 Findings

**2.2.1 Multimodal Reviews Impact Purchase Decisions.** The majority of participants (97.74%, 173 participants) mentioned product review videos influence their purchase decisions. For instance, P8 mentioned how "... seeing the product used in real-life situations gives me the confidence to buy [it]". Review videos were also valued for

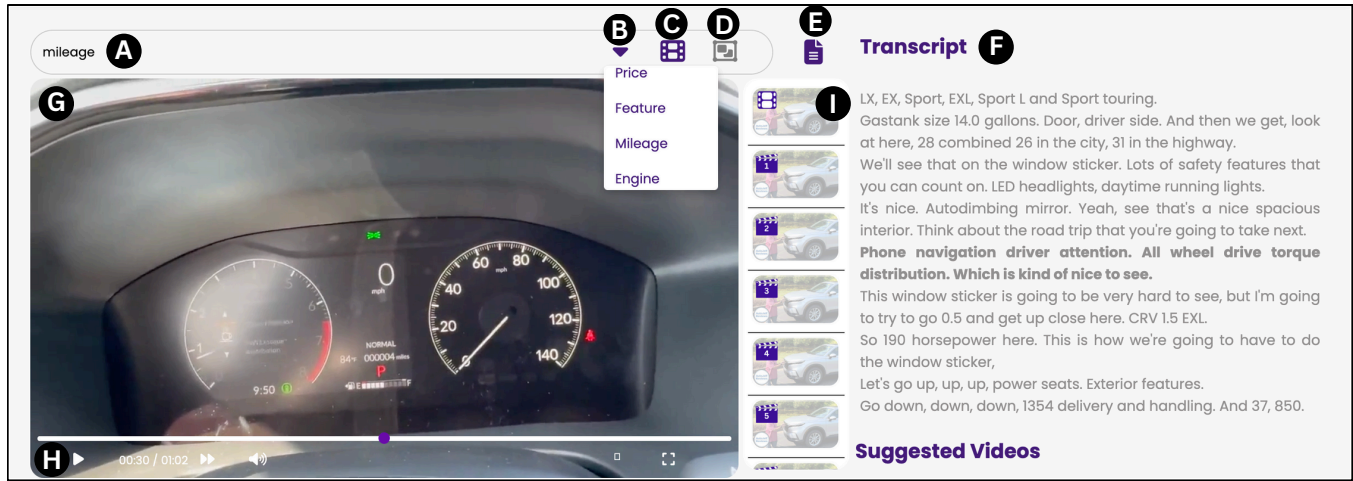
providing firsthand accounts of others' experiences with a product. P15 said, "*The videos showed features and performance information [that] I need to decide whether the product is really what I am looking for*". Similarly, P81 noted, "*Those videos make me reconsider whether the investment is worth it*". Some participants also mentioned how others' comments on the video helped build trust, "*Sometimes, I trust the feedback from other viewers more than the actual reviewer because it feels more genuine.*" (P42)

**2.2.2 Video Length Affect Watch Experiences.** Our participants demonstrated divergent inclinations toward the length of product review videos. While the majority (83.05% 147 participants) preferred videos shorter than 15 minutes for quick decision-making, 49.71% (88) participants thought longer videos provided more comprehensive insights. Participants pointed out frustration with long videos when critical information was difficult to find. P23 said, "*Sometimes the reviews [videos] are too long and I don't know where the information is that I want*", whereas P54 said, "*Lengthy videos take a long time to arrive at the feedback part.*" In contrast, some participants mentioned how short videos "*lack information*" (P19) and "*do not have all relevant information [about products]*" (P88). Many participants found summaries useful in separating the pros and cons of a product. P27 said, "*... quick recaps about the pros and cons are very helpful for deciding if I should buy the product*". P68 also preferred, "*... videos with summaries or searchable content to find specific details quickly.*"

**2.2.3 Strategies to Navigate Review Videos.** 62.14% (110) participants indicated that they skip or fast-forward sections of videos because the content is too long. Others mentioned using chapters, timestamps, and speed controls to find the necessary information. They emphasized the need for navigational aids to improve their experience. P52 said, "*If the video is structured properly with timestamps or sections, I can go straight to the part I need.*" P33 also included "*Bookmarks allow me to revisit specific details I need later, without re-watching the entire video*". Participants also mentioned different playback speeds can increase efficiency. P160 mentioned, "*I often watch videos at 1.5x speed to get through the information more quickly.*" P16 noted, "*To save time, I use fast forward or playback speed adjustment to get through videos faster.*"

**2.2.4 The Need for Comprehensive Product Information.** Participants sought information including quality, applicability, specific dimensions, compatibility, and features in review videos. They mentioned how "*detailed information is important*" (P120) and "*without proper specs, it's hard to judge whether a product meets expectations.*" (P40) P53 mentioned, "*I trust reviews that show how the product performs.*" Moreover, participants highlighted how transcripts can help in accessing information easily. P21 indicated, "*A transcript would let me skim the content rather than watch the full video*", while another (P135) explained, "*Reading a quick overview or transcript will save me so much more time than having to watch the full video.*" These features were particularly useful for consumers who value efficient information gathering.

**2.2.5 Establishing Credibility and Trust.** We found credibility to be a major factor in purchase decisions, as participants highlighted how authenticity and transparency impact trust in product review videos. P22 explained, "*Product review videos drive purchase decisions*



**Figure 1: Custom summary view.** A) Users can type their query in the search bar. B) 'Frequent Searches' has the suggested queries based on prior queries. C) Pressing "Generate Custom Summary" creates and shows the personalized video summary. D) 'Overlay' button gives the highlight in the seek bar where the video segments are within the original video seek bar. E) Hide/Show Transcript button is used to show or hide the transcript as per the user preference. F) An interactive transcript section to read and navigate the video. G) The Video player section displays the summarized video. H) The Video Control section has play/pause, next, volume control, speed, and full-screen controls. I) The clips section provides the segmented video clips along with the summarized video clip.

since buyers can obtain detailed information, clear illustrations, and firsthand accounts of users' experience." P32 said, "I have watched so many reviews about different products before purchasing my phone. That enables me to compare different devices and assess what best suits my needs." However, participants also raised concerns about the trustworthiness of reviews as P164 commented, "The reviewers might be biased because of sponsorships or collaboration with the company, which may lead to a biased review that may not fairly represent views about the product."

### 3 Synoptic

Based on the survey responses, we designed and developed Synoptic, a video player that allows users to find information from product review videos using custom queries. Synoptic provides two viewing options: (1) a customized summary view that provides a summary video based on users' query, and (2) an overlay view that plays the video with overlays highlighting parts that contain content people searched for. In addition, Synoptic also provides an interactive transcript to help users navigate through the content. In this section, we describe Synoptic in detail.

#### 3.1 Video to Text Transcription and Image Frame Generation

First, we isolate the video and audio data. We used MoviePy [49] to extract audio from the input video file and generated a Web Video Text Tracks (WebVTT)-based subtitle/caption file. Then, we segmented the audio using Pydub [11] into smaller segments based on the interval between prominent audio signals and lulls. This enabled us to identify meaningful audio cues from the review video, reduce transcription errors, and better organize important parts of the review without losing information. Individual audio segments

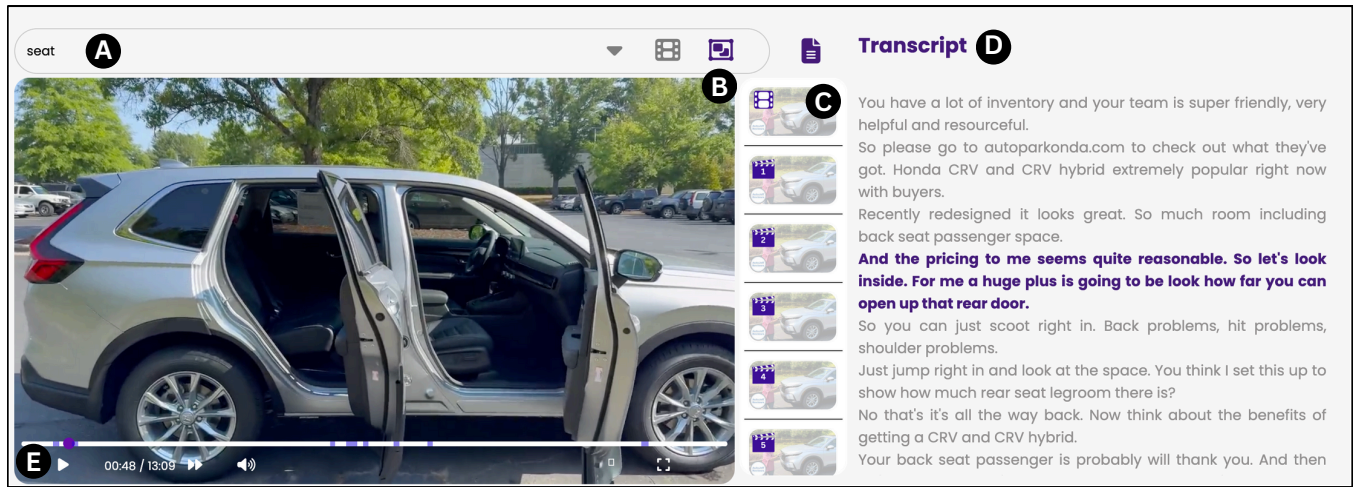
were then used to generate transcriptions using Google's Speech API and Speech Recognition Library [7]. The transcript was then split into individual sentences and annotated with timestamps. We saved the sentence-level transcript of the review video in a VTT file for additional processing.

We process the video files to extract key frames where a noticeable change occurs between consecutive frames. Using the OpenCV library [20], we read each frame of the video and calculate the structural similarity index (SSIM) [12] between consecutive frames to detect significant differences. If the SSIM score is less than the threshold value (0.5) [51], the frame and its timestamp are stored. This way, we isolate only the meaningful frames among all video frames.

#### 3.2 Custom Summary Generation

**3.2.1 Generating Custom Summary.** We developed a personalized multimodal summarization pipeline (Algorithm 1) to extract and process video clips based on the input query passed by the user. To generate the custom summary, we used a combination of GPT-4 [2] and Vision-and-Language Transformer (ViLT) [16]. The pipeline starts with extracting the search query and generating text embedding using GPT-4. These embeddings are then used to check if any cached entries match the user query. If there is a match, the relevant video clips that constitute the summary are combined to create the summary video and rendered to the user.

If the user query does not match any cached entries, a new summary generation cycle begins. First, we use the query text to identify segments within the transcript of the video that match the query embeddings using GPT-4. In parallel, we performed visual question answering on the extracted key frames to identify if the image frame contains information that is pertinent of the



**Figure 2: Overlay view.** A) Search bar for querying the video. B) The overlay button annotates the seek bar with the relevant video segments corresponding to the query. C) The clips section provides the segmented video clips along with the summarized video clip. D) The synchronized transcript E) The highlighted seek bar shows the positions of relevant video segments.

search query. Then, we check if the image frames with relevant information match the previously identified transcript segments. A match between the text segments and image frames reinforces that the segment in question indeed contains information that relates to the search query. We set a threshold of 80% between the text transcript and the key image frame to be considered a match. We used thresholding based on prior work that suggests it provides the best compromise between the relevance of image content and the flexibility of the matching process using the ViLT model [16]. A lower threshold could lead to the inclusion of unrelated or only remotely related visual content, whereas a higher threshold could mean that useful visual data is omitted. We ran several tests to tune our approach and we found that 80% was the best threshold limit for our system. If there are no such matches found, we prioritize the image frame over the text segment, due to images containing information that might be desired but relevant information was not uttered during the speech. For example, for the search query “color” on a product review of a car, we identify an image frame where the car is shown in full to understand its body color, even though the narrator did not mention color specifically. In these cases, we create an additional segment with the frame at its center. The interval of this segment is based on the length of the corresponding transcript segment as generated in the VTT file.

After identifying all relevant video segments, we merged them together using FFMPEG [22]. We added a fade-in fade-out effect between each segment to reduce the abrupt switching to the next segment, which has been shown to incur confusion and cognitive load when viewing video summaries [40]. The summary video is then cached alongside the query embeddings so that if other users search the video with similar queries, the video segments can be used in subsequent custom summaries. Finally, the summarized video is rendered for the users.

### 3.3 User interface and interaction

The synoptic user interface provides a search bar to input user queries (Fig. 1 (A)), followed by a dropdown button (Fig. 1 (B)), which can be used to search the video based on frequently searched keywords. This is followed by two view options - custom summary view (Fig. 1 (C)) and overlay view (Fig. 1 (D)). The last toggle option can be used to enable and disable interactive transcripts (Fig. 1 (E)). All of these options have interactive tooltips that provide descriptions when hovered over. The video player (Fig. 1 (G)) and the linked transcript (Fig. 1 (F)) are then rendered in juxtaposition. The video player has control options (Fig. 1 (H)) for seek, audio, and skimming for users. The custom summary view is designed to provide the users concise and relevant information regarding their queries. On the other hand, overlay view allows users to watch the full video with a highlighted seek-bar with the relevant portions of the video according to their queries.

**3.3.1 Custom Summary View.** The custom summary view enables users to generate custom summaries based on their input queries and view them. Users can type in their keywords (Fig. 1(A)) or select from the frequently used keyword dropdown to start their search process (Fig. 1(B)). If the user searches for information using frequent keywords, Synoptic searches for relevant content stored in the cache. If any relevant clips are found, they are stored and rendered, reducing summary generation time. On the other hand, if the searched keyword is new, 1 is initiated and a new custom summary based on the keyword search is generated. A loading wheel is rendered on the screen while the custom summary is being processed. The newly created clips are cached for later queries. The summary video is then played in the video player (Fig. 1 (G)). The individual clips are shown in a sidebar through the toggle option (Fig. 1 (I)). These clips are interactive and can be used to skip through the summary video.

**3.3.2 Overlay View.** Instead of rendering a shortened summary of the input review video, the overlay view renders the complete video

**Algorithm 1:** Generate Custom Summary

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```

Input: video_frames, vtt_file, video_path, output_folder,
        query, callback
Output: merged_video, valid_timestamps, output_folder

1 segments ← Parse-VTT(vtt_file); /* Parse subtitles
   into timestamps and text */
2 embeddings ← Generate-Embeddings(segments.text);
   /* Get GPT-4 embeddings for text segments */
3 query_embedding ← Generate-Embedding(query);
   /* Get GPT-4 embedding for the query */
4 matches ← Find-Top-Matches(query_embedding,
   embeddings); /* Identify relevant segments for
   the query */
5 results ← ∅; /* Initialize results to store
   timestamps */
6 foreach image ∈ video_frames do
7   pred ← Analyze-Image(image, query); /* Use ViLT
   model to perform VQA to see if the frame
   contains relevant information */
8   if pred.similarity ≥ 0.8 then
9     timestamp ← Generate-Timestamp(image,
       matches); /* Determine timestamp for the
       match */
10    Add timestamp to results;
11 valid_timestamps ← Sort-Timestamps(results);
   /* Organize timestamps in chronological order */
12 Extract-Clips(video_path, valid_timestamps,
   output_folder, callback); /* Extract video clips */
13 merged_video ← Merge-Clips(output_folder); /* Merge
   extracted clips */
14 return merged_video, valid_timestamps, output_folder;

```

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but adds visual cues that highlight the parts of the videos that match custom user queries. Similar to the custom summary generation, the process starts with users searching for keywords using the search bar (Fig. 2 (A)) and then click the overlay button (Fig. 2 (B)). The custom summary generation process follows (Algorithm 1). Once the custom summary is generated, the player plays the complete video. However, the seek bar now highlights the parts of the video that contain information relevant to the user’s search query. Users can click on the highlighted options to traverse the video or use the clips sidebar to navigate to the information they need (Fig. 2 (E)).

**3.3.3 Transcript.** Synoptic presents an interactive and linked transcript that are rendered alongside the video (Fig. 1 (F)). The transcripts are responsive and matches with the custom summary, making the output summary multimodal (video and text). This transcript is designed to scroll in parallel with the video so that the text corresponding to the video is shown to the user as the video is being played. The scrolling was implemented to enable users to see the video and glance at the transcript at the same time. The synchronization operates with the help of an event listener on the video player, which follows the current time and emphasizes the

fragments of dialogue or narration being said at the moment. The transcript can be shown or hidden using a toggle button (Fig. 1 (E)).

## 4 Evaluation

We conducted an initial evaluation of Synoptic with 40 participants who used our system to watch product reviews and make purchase decisions. From their feedback, we found that Synoptic enabled them to personalize and customize their product review video-watching experience, identify information efficiently, and make confident purchase decisions.

### 4.1 Participants and Procedure

We recruited 40 participants through Prolific. Among the participants, 25 used he/him, and 15 used she/her pronouns. Their average age was  $36 \pm 10.98$  years, ranging from 18 to 65 years. The majority (92.11%, 37 participants) watched product review videos before purchasing. All participants were from the US or UK and were proficient in English.

We selected two products — Sport Utility Vehicle (SUV) and Smartphone, due to their popularity [18, 19]. We selected three product review videos for each category from YouTube based on their views and like/dislike ratios. The average length of the videos was  $12.51 \text{ minutes} \pm 1.10 \text{ minutes}$ . We randomly assigned 20 participants to each product category. First, the participants filled up a pre-study questionnaire about their demographics and purchase decision-making practices. Then they watched all three videos using our tool and make a purchase decision based on their exploration. Participants spent an average of  $37.86 \pm 18.55$  minutes using Synoptic to make their purchase decisions. This was followed by a post-study questionnaire where they expressed their experiences of using Synoptic, their decision-making process, and suggestions for improving the system. Each participant was compensated with USD \$20 per hour.

### 4.2 Findings

**4.2.1 Personalized Video Summaries Enabled Efficient Information Gathering.** Our participants’ feedback suggests their preferences for personalization mechanisms, such as keyword search, personalized summaries, and overlays that allowed them to tailor long-form content to their needs. These enabled participants to cut through irrelevant details and reach the information most important to them. One participant (P40) mentioned, “*It was easy to just type in what I wanted to see in the video*” showing how targeted sections of a video were easily identified by them. Another participant (P9) said, “*Custom summary saved me time from needing to skip around the video*”. P37 appreciated the specificity of the summaries: “*I needed to know just the price, battery life, and colors in which each phone was available, so I used the custom summary to compile just those points*.” However, some participants raised concerns around “*missing information*” (P21). For instance, P35 mentioned, “*Sometimes the summary tool skipped parts of the video that I felt were important*.”

**4.2.2 Effective Navigation Enhanced User Experience.** Based on participant feedback, they found various navigation options useful for finding relevant information. For instance, the overlay view was heavily praised for enabling quick access to specific content. P7 mentioned, “*Overlay feature made it easy to find important parts*

of the video quickly.” Others (P4) mentioned how the interactive transcript helped them navigate the video efficiently: “I liked how the transcripts followed the video in real-time with relatively high accuracy of the words being said.” Another (P17) noted, “Skimming through the transcript helped me decide whether the video has the information I need.” The overlay was particularly preferred over the customized summary by participants who valued detailed information. For instance, P10 mentioned, “Long-form reviews go into more detail about the product’s features, pros and cons, and real-life usage... I prefer my own understanding of the video instead of using a summary.”

**4.2.3 Synoptic Facilitates Agency by Prioritizing User Preferences.** From participants’ responses, we found that Synoptic enabled them to prioritize their preferences when interacting with the system and videos. Through the interconnected multimodal approach (customized videos, visual overlays, and text transcript), Synoptic allowed participants to approach product review video consumption, analysis, and decision-making at their own pace and agency. P32 mentioned, “I appreciate how synoptic can streamline information and make it easier to digest.” Referring to Synoptic’s interface, P32 mentioned, “I liked how it had everything in one convenient place.” Others including P24 mentioned the interconnectivity of components, “It was easy to read the video transcript and follow along, and I could jump to different sections of the video depending on the search I made (custom summary) or clicking on the transcript.” Although participants (P6) found the custom summaries being “...unbiased and informative.”, some participants (P38) simply ... don’t trust technology” or suspected that “it (the summary) might not be totally accurate depending on the content being summarized.”

## 5 Conclusion and Future Works

The findings from our initial evaluation suggest participants’ preference towards multimodal customized summaries of the product review videos. Participants appreciated the prioritization of their preferences when generating summaries and found Synoptic to be useful for maintaining user agency [57]. They also highlighted how the summaries allowed them to gather information efficiently to make purchase decision effectively. While participants were generally positive around Synoptic, some participants raised concerns about the automated summaries. These concerns arose from a lack of trust in technology, inaccuracy in summaries, and whether the summaries were contextually appropriate. These findings corroborate with prior works that highlight similar concerns around text summaries of product reviews [27, 30, 47].

In its current state synoptic generated summary reviews based on a single summary. In the future, Synoptic could be improved to generate summary reviews across multiple videos, creating a parallel with current text summarization across product reviews [33, 58]. Further studies are required to investigate whether product review videos are J-shaped, similar to their textual counterparts [24, 25]. Future iterations of Synoptic could incorporate technologies that not only generate summaries across reviews but also present a balanced outlook on the product across reviews [30]. This might be beneficial in promoting trust in viewers and negating biased viewpoints toward products [45].

In the future, we will expand upon this study in two key ways. First, we will explore domains other than product reviews where custom query-based video summarization could be beneficial. For instance, in the education domain, Synoptic could allow viewers to explore and identify specific topics from long-form (often tens of hours long) tutorial videos. Additionally, such approaches can be beneficial in meeting discussion analysis where meeting videos are recorded for analysis [15, 31]. Second, we will increase the scalability by developing a widget version of Synoptic, which will allow the generation of customized video summaries on embedded videos online. While further investigation and longitudinal studies are needed to assess the viability of multimodal summarization systems, given the recent shift towards multimedia content, we posit that such systems will be beneficial across several domains. We will also conduct more rigorous quantitative experiments to compare Synoptic with existing text-based approaches.

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## A Themes

### A.1 Needfinding Survey Themes

The need finding survey was conducted with 177 participants. We analyzed the survey responses using thematic analysis. The themes, their descriptions, and example codes are presented here.

**Table 1: This table shows the themes based on the survey analysis. The table also includes example codes linked to the themes, along with an explanation of each.**

| Themes                              | Example Codes                                                                                                                                                                                                         | Descriptions                                                                 |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Shaping Purchase Decision           | helpful in decision-making, good reviews, incorrect information, lack information, more information, more explanation, comparison, informed choices, visual demonstrations, pros and cons, user experiences, feedback | Elements of product review videos that assist in making a purchase decision. |
| Preferences for Video Duration      | prefer short videos, longer for complex products, shorter for simple products, summary, one line conclusion, optimal length 5-15 mins, avoid filler                                                                   | Preferences for product review video duration.                               |
| Strategies to navigate and interact | timestamp markers, clear sections, easy access, fast-forwarding, skipping, playback speed adjustment                                                                                                                  | Strategies that help viewers navigate through a video and interact better.   |
| Product information                 | detailed specs, real-world usage, durability tests, comparisons, value for money, unboxing, skimming transcript                                                                                                       | Types of information viewers prioritize in product review videos.            |
| Presentation Style                  | not overly animated, boring, straightforward, engaging visuals, clear language, avoid fluff, logical structure, concise contents                                                                                      | Elements of video that enhance the presentation.                             |
| Credibility and Trust               | unbiased opinions, transparency about sponsorships, reviewer expertise, authentic feedback                                                                                                                            | Refers to qualities that establish trust.                                    |

### A.2 Tool Evaluation Themes

Our initial evaluation of Synoptic was conducted with 40 participants. We analyzed their pre- and post-task surveys with using thematic analysis. The themes, their descriptions, and example codes are presented here.

**Table 2: This table shows the thematic analysis of initial user feedback. The table has the example codes and descriptions of those codes.**

| Themes                 | Example Codes                                                                                  | Descriptions                                                               |
|------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Customization Features | keyword search, custom summaries, custom overlays                                              | Custom features that allow users to find specific information efficiently. |
| Depth and Brevity      | balance between depth and efficiency, clear timestamps, optional summaries                     | How platforms can cater to both short- and long-form content preferences.  |
| Efficient Navigation   | searchable content, timestamps, transcripts, chapter markers, visual cues, overlay features    | Tools and features for video navigation without watching the entire video. |
| User Priorities        | efficient navigation, accurate summaries, seamless interface                                   | Areas users feel should be prioritized.                                    |
| Trust factors          | unbiased reviews, untrust technology, visual aids, build trust, more relatable and trustworthy | Key factors that enhance user trust in product reviews.                    |